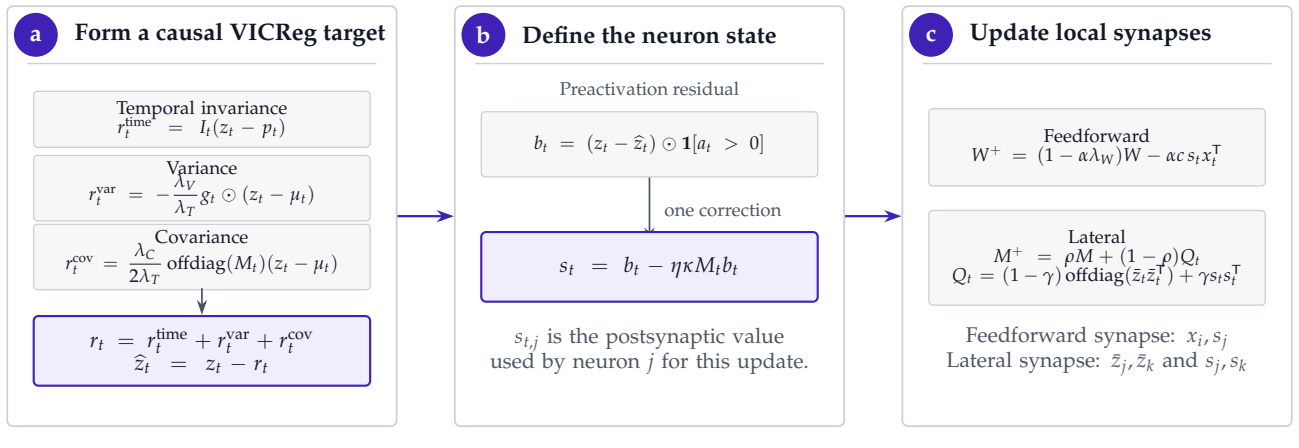
a
Form a causal VICReg target
Temporal invariance
$r_t^{\text{time}} = I_t(z_t - p_t)$
Variance
$r_t^{\text{var}} = -\frac{\lambda_V}{\lambda_T} g_t \odot (z_t - \mu_t)$
Covariance
$r_t^{\text{cov}} = \frac{\lambda_C}{2\lambda_T} \text{offdiag}(M_t)(z_t - \mu_t)$
$r_t = r_t^{\text{time}} + r_t^{\text{var}} + r_t^{\text{cov}}$
$\hat{z}_t = z_t - r_t$
b
Define the neuron state
Preactivation residual
$b_t = (z_t - \hat{z}_t) \odot \mathbf{1}[a_t > 0]$
one correction
$s_t = b_t - \eta\kappa M_t b_t$
$s_{t,j}$ is the postsynaptic value used by neuron j for this update.
c
Update local synapses
Feedforward
$W^+ = (1 - \alpha\lambda_W)W - \alpha c\, s_t x_t^\top$
Lateral
$M^+ = \rho M + (1 - \rho)Q_t$
$Q_t = (1 - \gamma)\,\text{offdiag}(\bar{z}_t \bar{z}_t^\top) + \gamma s_t s_t^\top$
Feedforward synapse: x_i, s_j
Lateral synapse: $\bar{z}_j, \bar{z}_k$ and s_j, s_k
One observation at a time
detached state
no error signal from a later layer